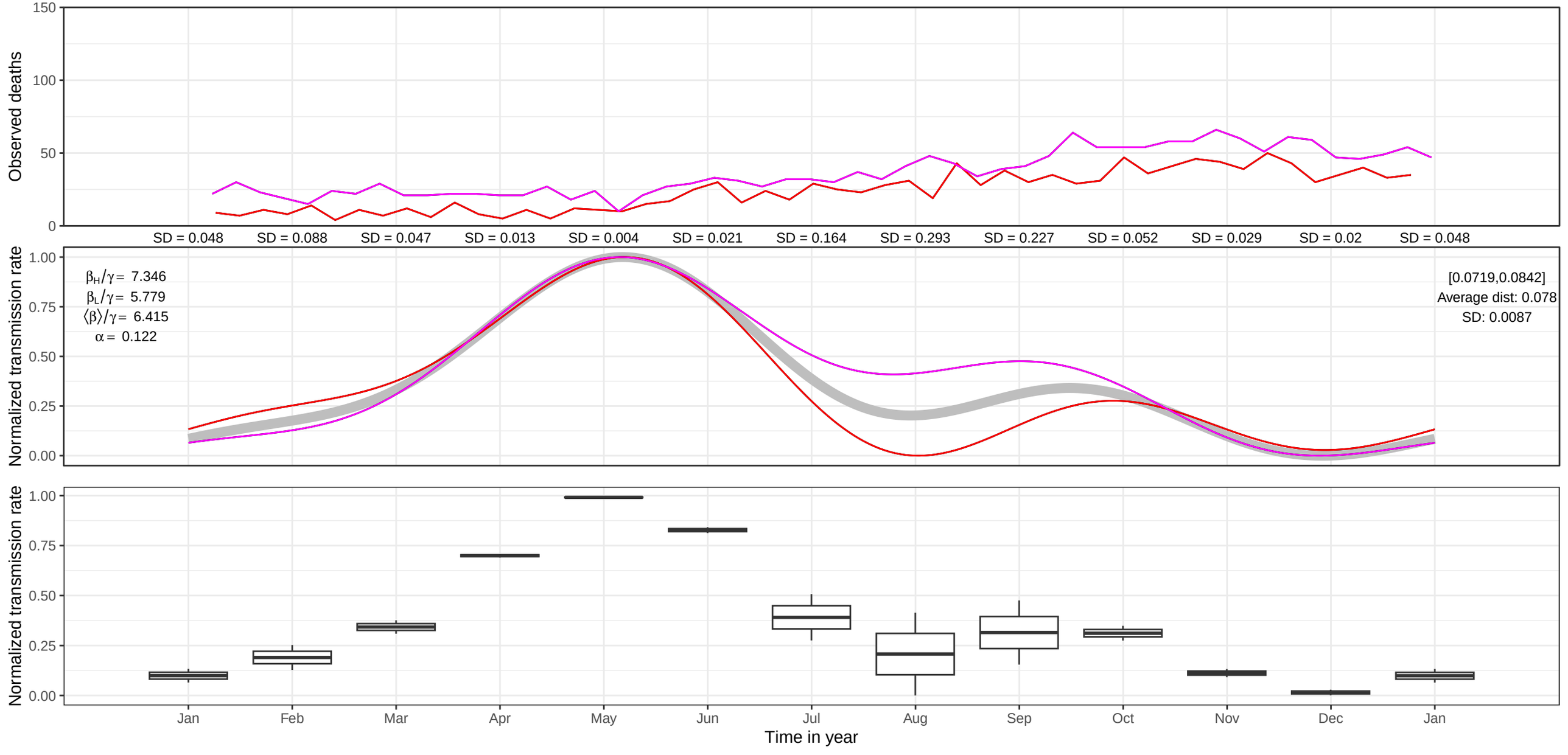
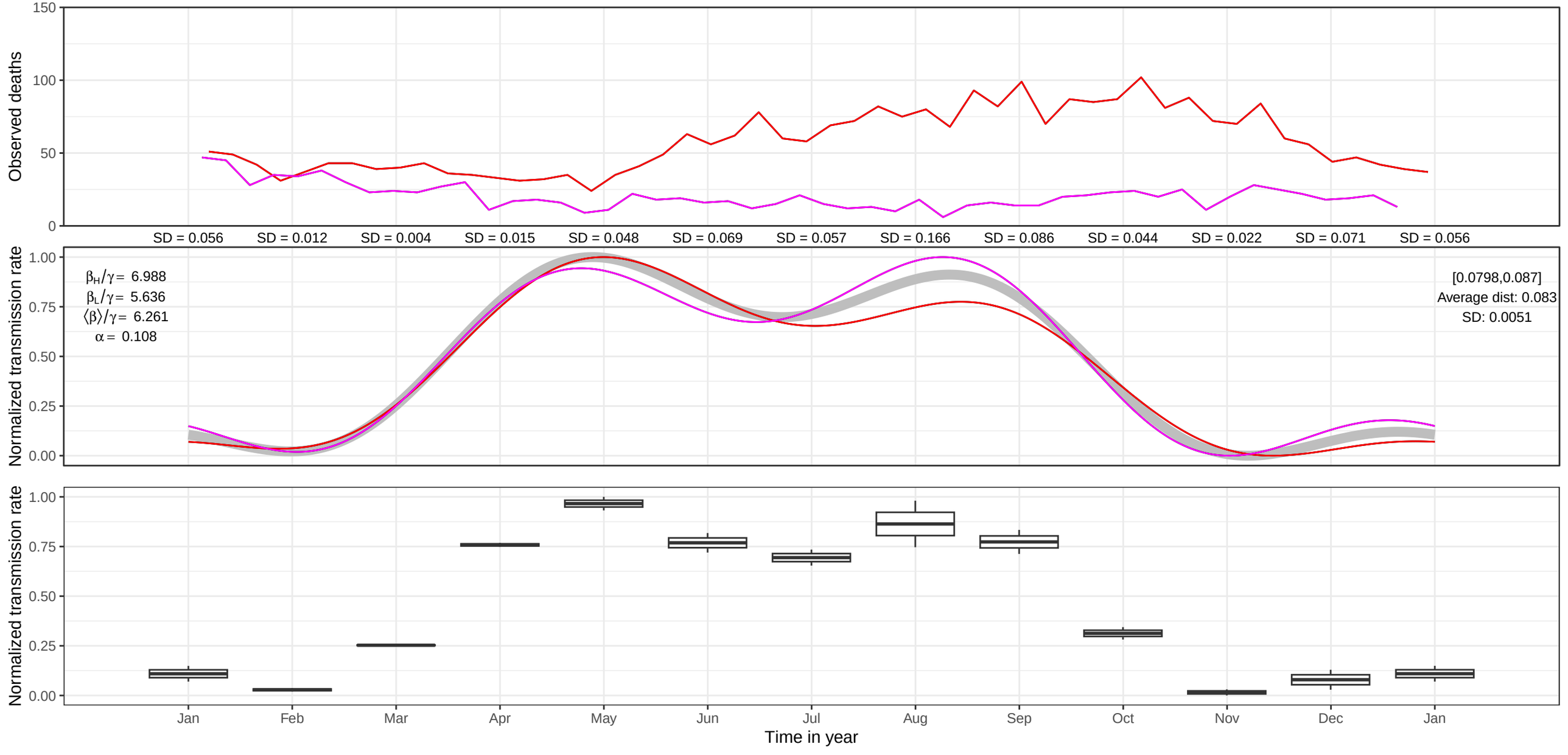
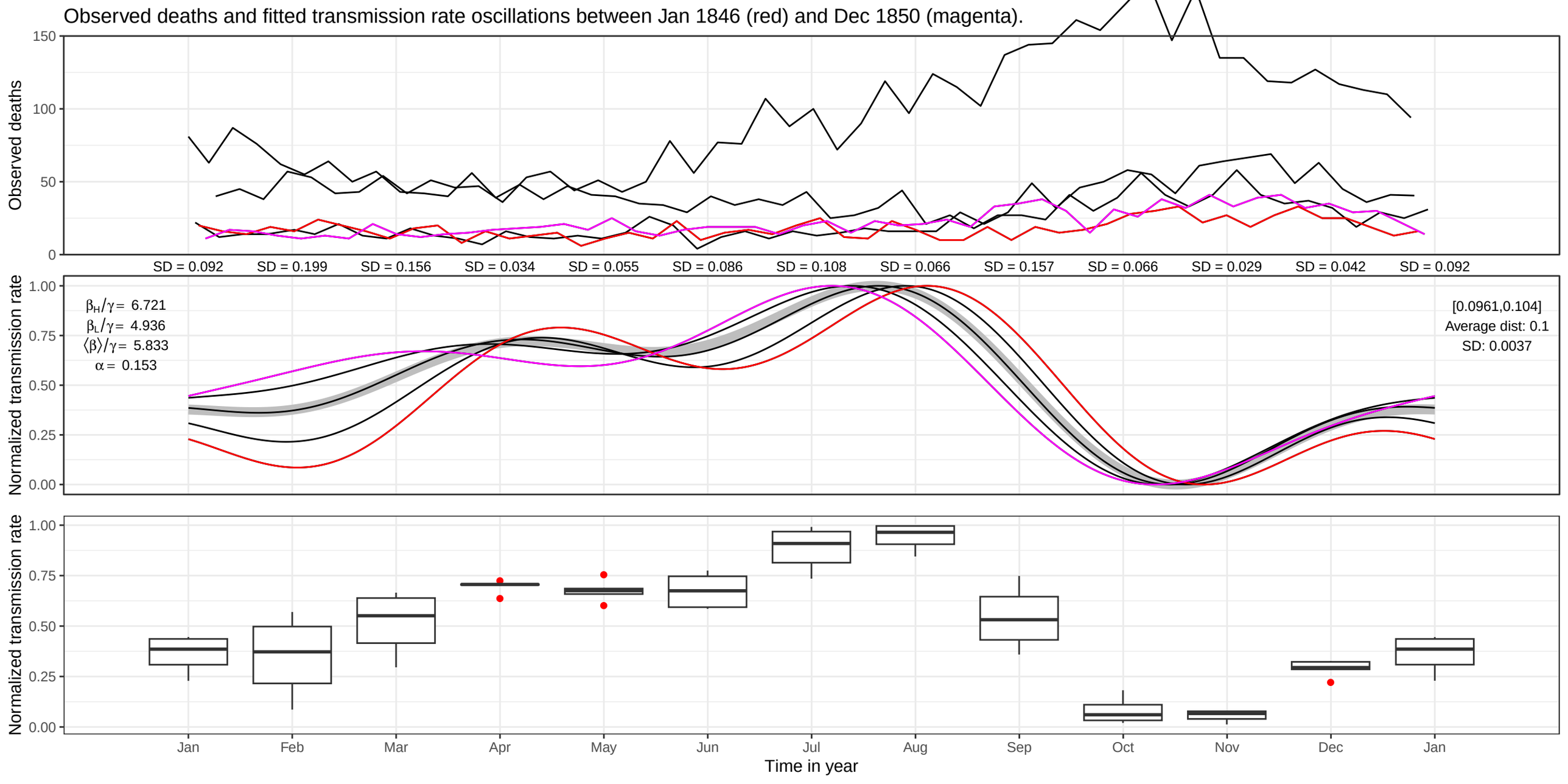
Observed deaths and fitted transmission rate oscillations between Jan 1842 (red) and Dec 1843 (magenta).

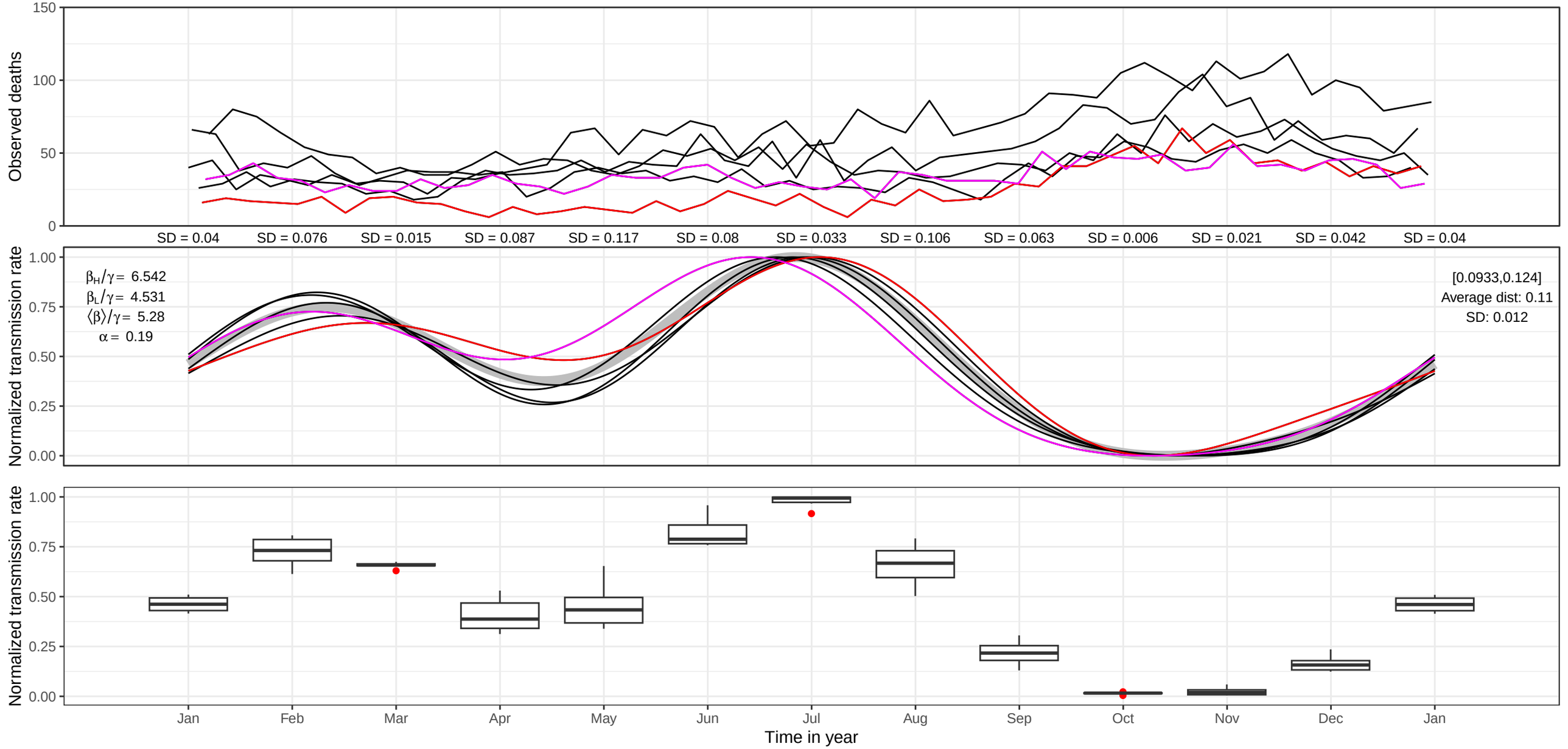


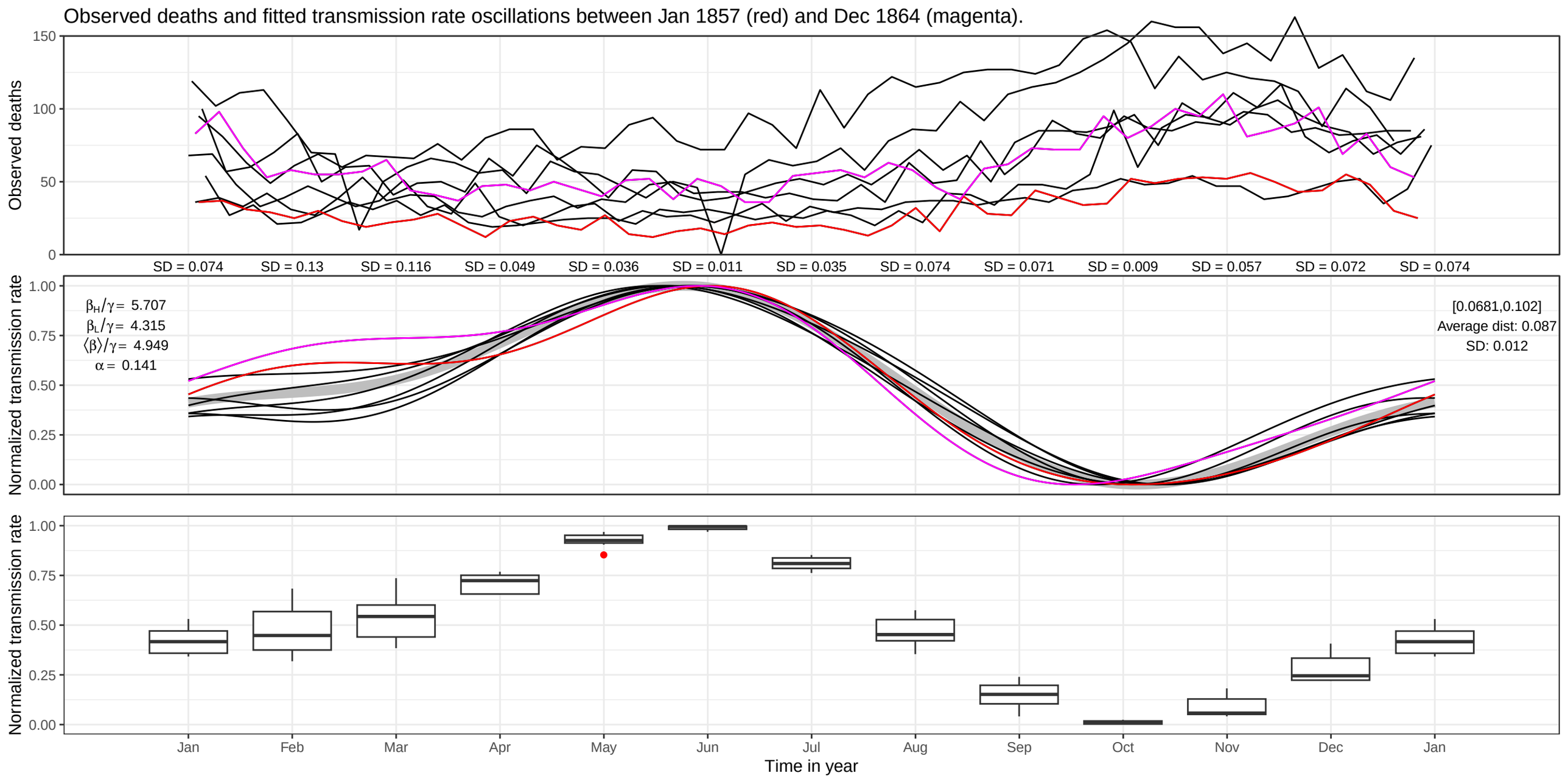
Observed deaths and fitted transmission rate oscillations between Jan 1844 (red) and Dec 1845 (magenta).

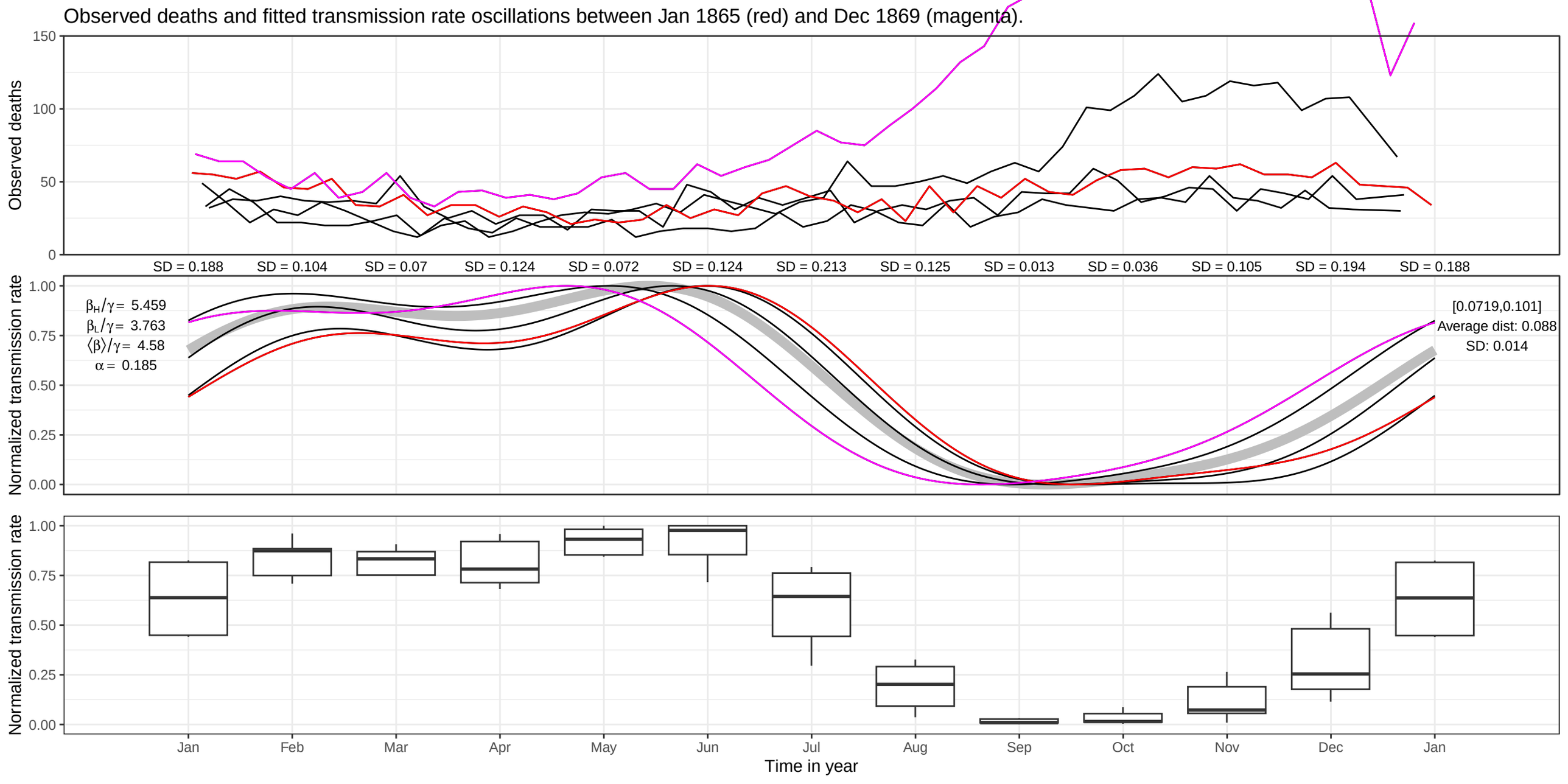


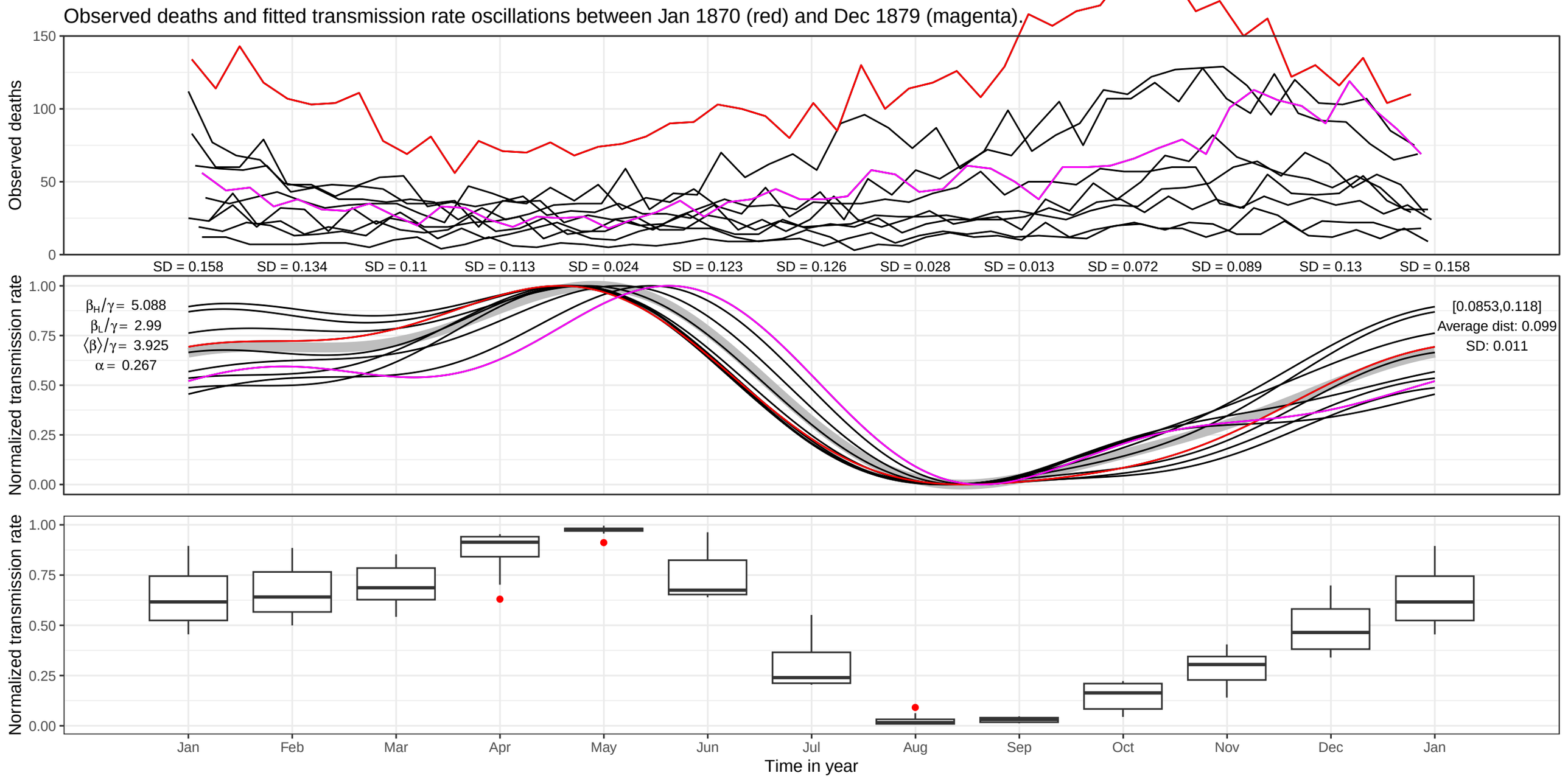

Observed deaths and fitted transmission rate oscillations between Jan 1846 (red) and Dec 1850 (magenta).
Observed deaths
150
100
50
0
SD = 0.092
SD = 0.199
SD = 0.156
SD = 0.034
SD = 0.055
SD = 0.086
SD = 0.108
SD = 0.066
SD = 0.157
SD = 0.066
SD = 0.029
SD = 0.042
SD = 0.092
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
βH/γ= 6.721
βL/γ= 4.936
⟨β⟩/γ= 5.833
α= 0.153
[0.0961,0.104]
Average dist: 0.1
SD: 0.0037
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

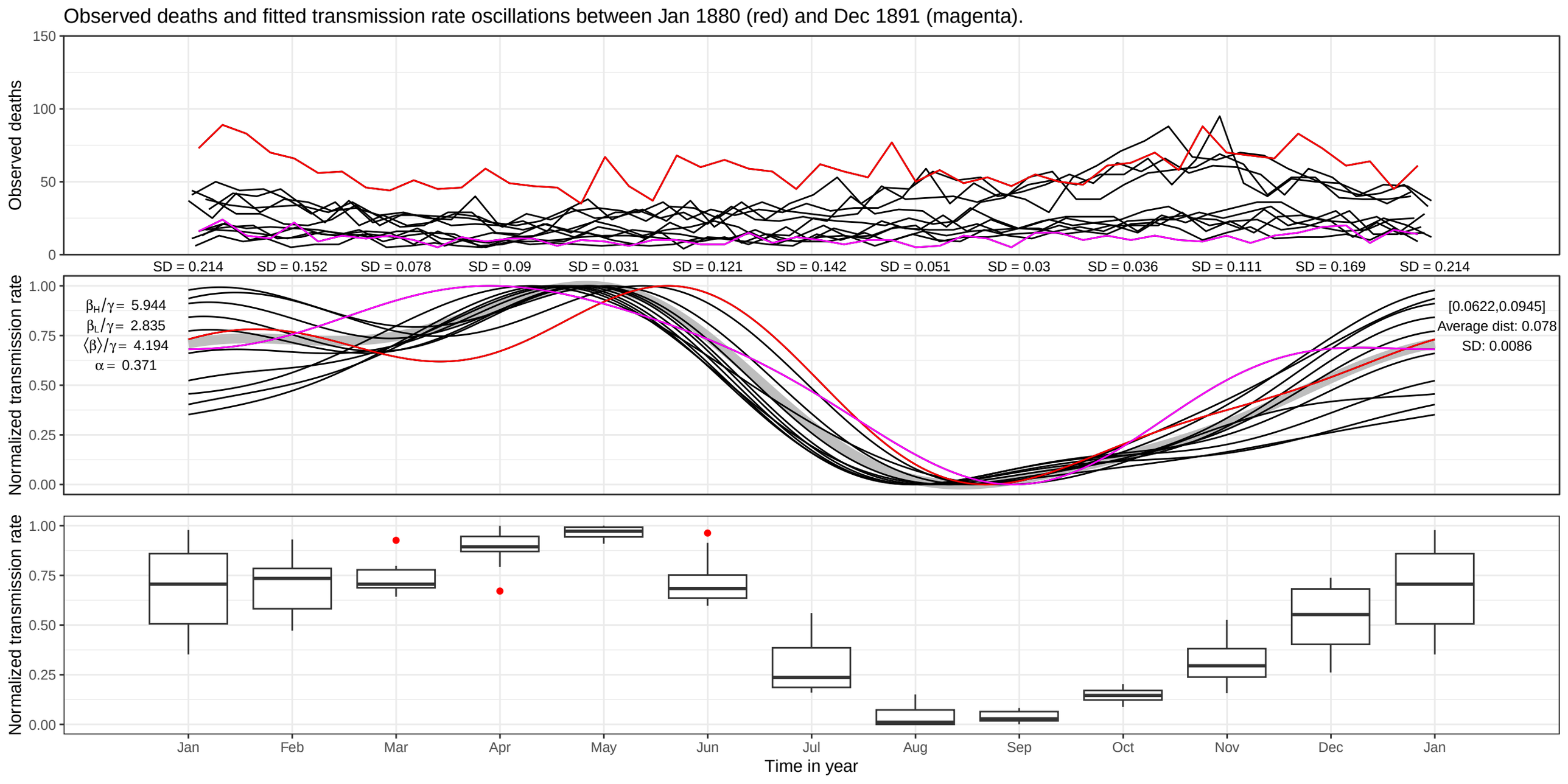
Observed deaths and fitted transmission rate oscillations between Jan 1851 (red) and Dec 1856 (magenta).



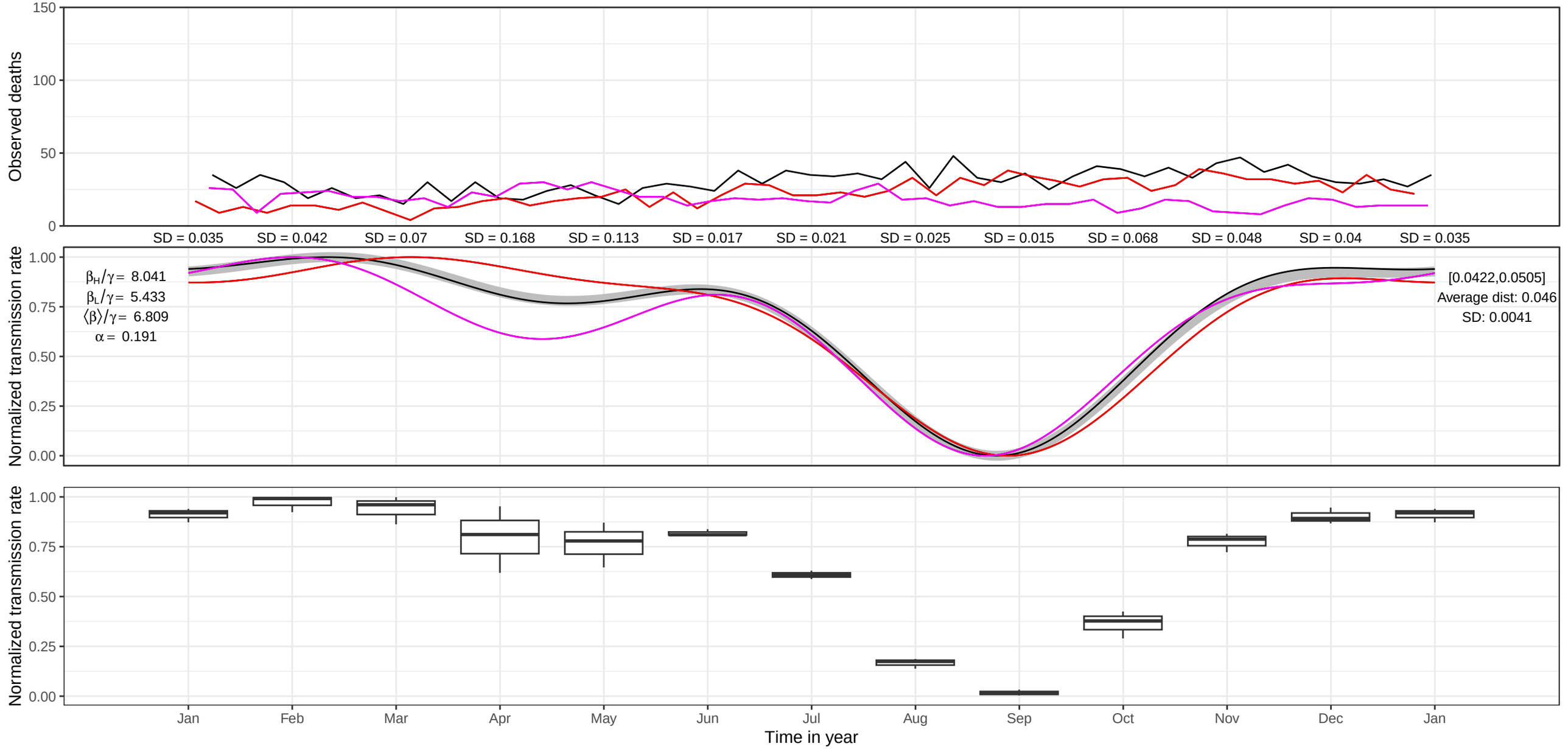

Observed deaths and fitted transmission rate oscillations between Jan 1857 (red) and Dec 1864 (magenta).
Observed deaths
150
100
50
0
SD = 0.074
SD = 0.13
SD = 0.116
SD = 0.049
SD = 0.036
SD = 0.011
SD = 0.035
SD = 0.074
SD = 0.071
SD = 0.009
SD = 0.057
SD = 0.072
SD = 0.074
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
$\beta_H/\gamma =$ 5.707
$\beta_L/\gamma =$ 4.315
$\langle\beta\rangle/\gamma =$ 4.949
$\alpha =$ 0.141
[0.0681,0.102]
Average dist: 0.087
SD: 0.012
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year


Observed deaths and fitted transmission rate oscillations between Jan 1865 (red) and Dec 1869 (magenta).
Observed deaths
150
100
50
0
SD = 0.188
SD = 0.104
SD = 0.07
SD = 0.124
SD = 0.072
SD = 0.124
SD = 0.213
SD = 0.125
SD = 0.013
SD = 0.036
SD = 0.105
SD = 0.194
SD = 0.188
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
$\beta_H/\gamma =$ 5.459
$\beta_L/\gamma =$ 3.763
$\langle\beta\rangle/\gamma =$ 4.58
$\alpha =$ 0.185
[0.0719,0.101]
Average dist: 0.088
SD: 0.014
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

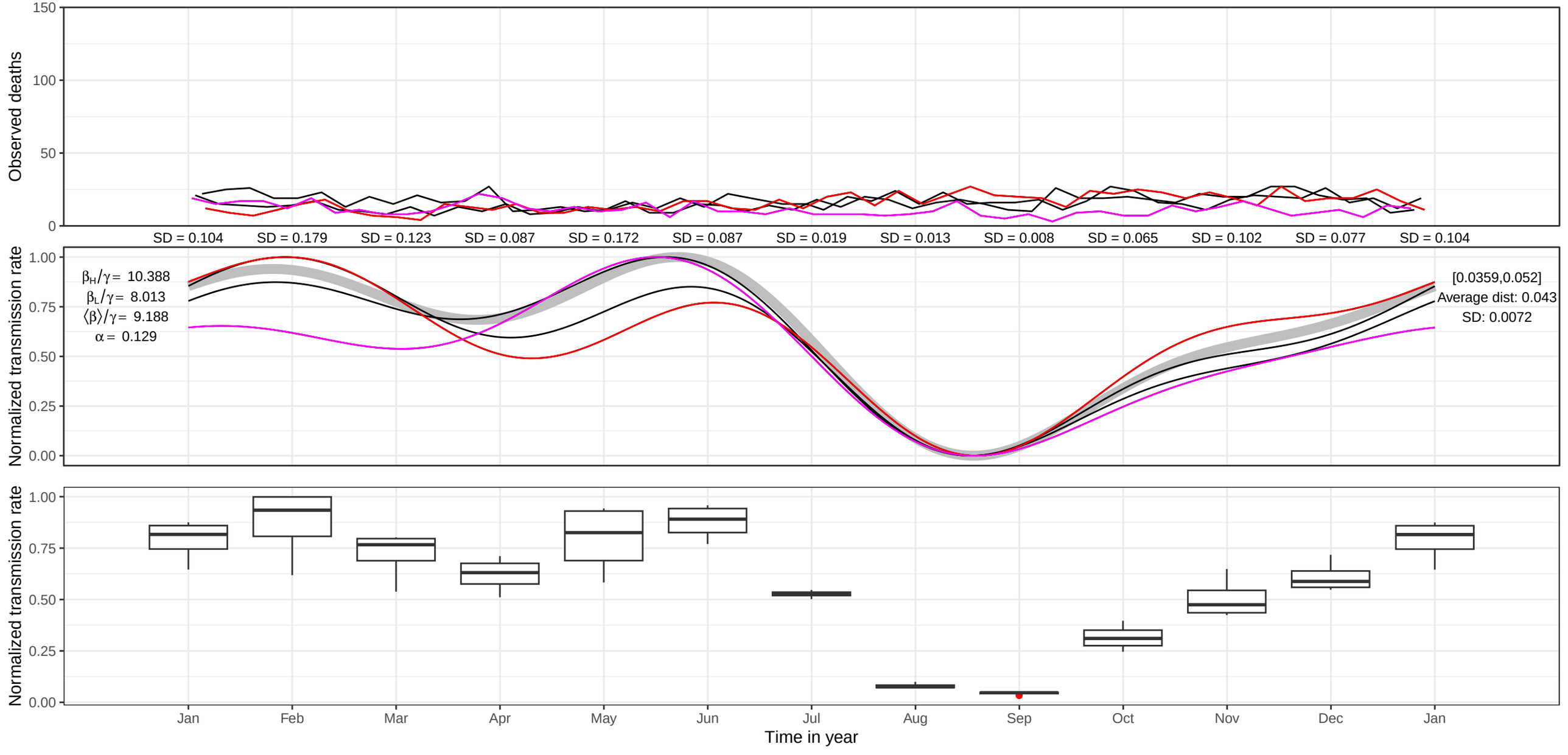
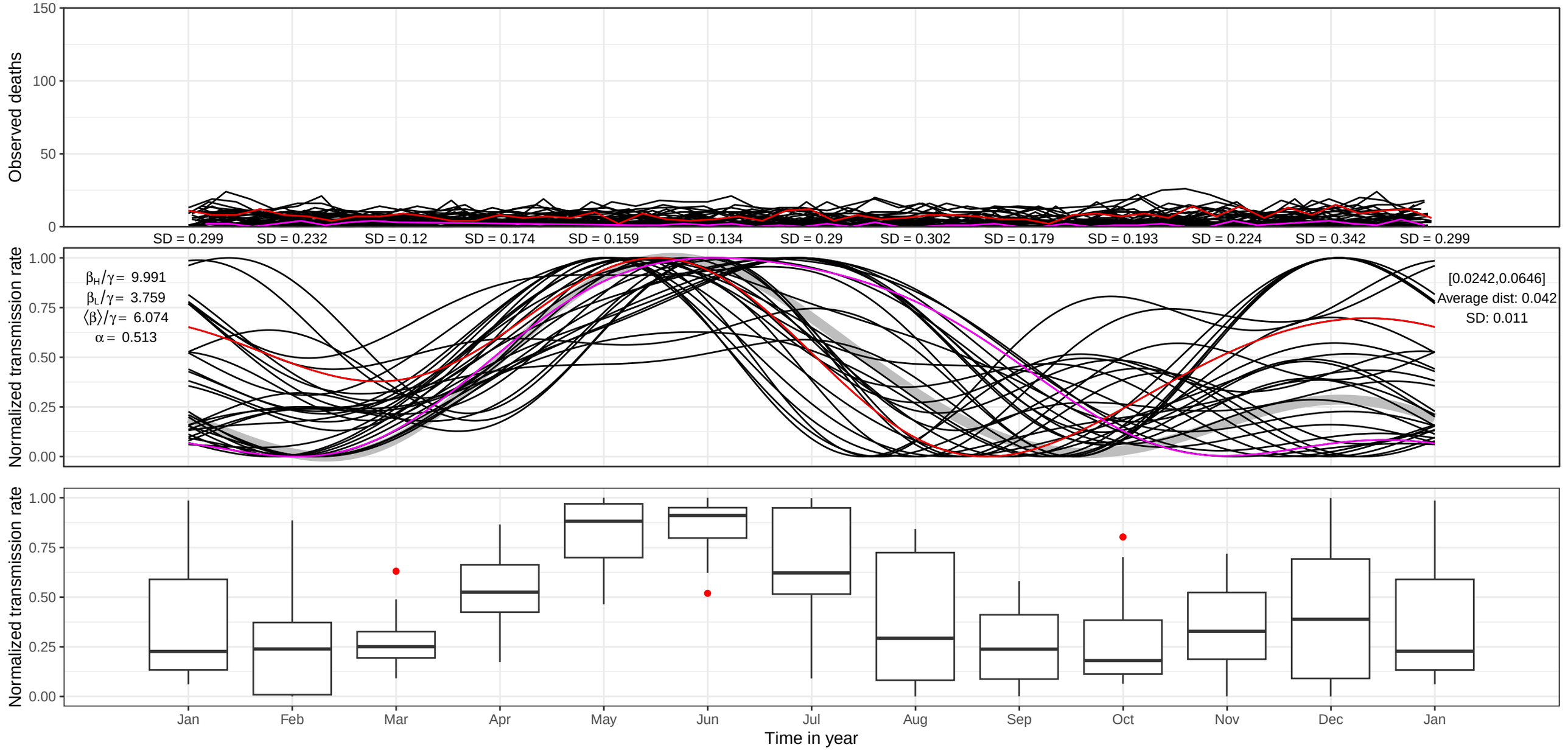

Observed deaths and fitted transmission rate oscillations between Jan 1870 (red) and Dec 1879 (magenta).
Observed deaths
150
100
50
0
SD = 0.158
SD = 0.134
SD = 0.11
SD = 0.113
SD = 0.024
SD = 0.123
SD = 0.126
SD = 0.028
SD = 0.013
SD = 0.072
SD = 0.089
SD = 0.13
SD = 0.158
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
$\beta_H/\gamma =$ 5.088
$\beta_L/\gamma =$ 2.99
$\langle\beta\rangle/\gamma =$ 3.925
$\alpha =$ 0.267
[0.0853,0.118]
Average dist: 0.099
SD: 0.011
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year


Observed deaths and fitted transmission rate oscillations between Jan 1880 (red) and Dec 1891 (magenta).
Observed deaths
150
100
50
0
SD = 0.214
SD = 0.152
SD = 0.078
SD = 0.09
SD = 0.031
SD = 0.121
SD = 0.142
SD = 0.051
SD = 0.03
SD = 0.036
SD = 0.111
SD = 0.169
SD = 0.214
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
$\beta_H/\gamma =$ 5.944
$\beta_L/\gamma =$ 2.835
$\langle\beta\rangle/\gamma =$ 4.194
$\alpha =$ 0.371
[0.0622,0.0945]
Average dist: 0.078
SD: 0.0086
Normalized transmission rate
1.00
0.75
0.50
0.25
0.00
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Jan
Time in year

Observed deaths and fitted transmission rate oscillations between Jan 1892 (red) and Dec 1894 (magenta).



Observed deaths and fitted transmission rate oscillations between Jan 1895 (red) and Dec 1898 (magenta).



Observed deaths and fitted transmission rate oscillations between Jan 1899 (red) and Dec 1929 (magenta).